

Lab-4

Task-1 :

A	B	C	F(A, B, C)
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1

Truth Table

Task-4 :

Current	State	Next	State
D_1	D_0	Q_1	Q_0
0	0	0	1
0	1	1	0
1	0	1	1
1	1	1	1

for Q_0 :

$\overline{D_1} \overline{D_0}$	0	1
$\overline{D_1} D_0$	1	0
$D_1 \overline{D_0}$	1	1
$D_1 D_0$	1	1

$$Q_0 = \overline{D_0} + D_1$$

for Q_1 :

$\overline{D_1} \overline{D_0}$	0	1
$\overline{D_1} D_0$	0	1
$D_1 \overline{D_0}$	1	1
$D_1 D_0$	1	1

$$Q_1 = D_1 + D_0$$

Down Counter 8

D₁ D₀

0 0

0 1

1 0

1 1

Q₁ Q₀

0 0

0 0

0 1

1 0

Q_0	Q_1	
0	1	
0	0	
0	0	
1	0	

$$Q_0 = \overline{D_0} D_1$$

Q_0	Q_1	
0	1	
0	0	
0	0	
1	0	

$$\Rightarrow Q_1 = D_0 D_1$$